

Problem 15.19

Megan purchased a new car for \$18,000. She finances the entire purchase over 60 months at a nominal rate of 12% compounded monthly. Calculate Megan's monthly payment.

Solution

The nominal annual interest rate compounded monthly is 12%, so the monthly interest rate is

$$j = \frac{0.12}{12} = 0.01.$$

The number of monthly payments is

$$n = 60.$$

Let X be Megan's monthly payment. Since she finances the entire purchase, the present value of the monthly payments must equal \$18,000:

$$18000 = Xa_{\overline{60}|}.$$

Using

$$a_{\overline{n}|} = \frac{1 - (1 + j)^{-n}}{j},$$

we get

$$18000 = X \left(\frac{1 - (1.01)^{-60}}{0.01} \right).$$

Solving for X ,

$$\begin{aligned} X &= \frac{18000}{\frac{1 - (1.01)^{-60}}{0.01}} \\ &\approx 400.40. \end{aligned}$$

Hence, Megan's monthly payment is

$$\boxed{\$400.40}.$$